

COMPLETE PRESENTATION SPEECH SCRIPT

Week 2: Atmospheric Profile Retrieval with AI

SLIDE 1: Title Slide

Speech Script:

"Good morning/afternoon everyone. Today I'll be presenting my Week 2 project on building AI brains to guess the atmosphere. Over the past two days, I've been working on a fascinating problem: how to use artificial intelligence to reconstruct the full picture of the atmosphere from just a few measurements. Let me walk you through what I learned, the results I achieved, and how this technology can be applied in other fields."

SLIDE 2: What I Learned & How I Solved Problems

Speech Script:

"In this project, I built AI brains to guess the full atmosphere from just a few measurements. This is what we call an inverse problem - we have limited data and need to reconstruct the complete picture."

"I learned three main techniques. First, **PCA**, which shrinks big data down to just a few important numbers. Second, **1D-CNN**, which finds patterns in ordered numbers - like how the atmosphere changes with height. And third, **Residual MLP**, which uses something called skip connections to learn much better."

"Now, I faced some significant technical challenges along the way. Python 3.13 was too new for PyTorch, so I created a Python 3.12 environment. NumPy wouldn't work properly with PyTorch, so I downgraded to version 1.26.4. SciPy had version conflicts, so I installed version 1.11.4. And GitHub wouldn't accept my password, so I used a personal access token instead."

"The key lesson I learned from all of this is that **skip connections** really help AI brains learn much better. This insight became the foundation of my best-performing model."

SLIDE 3: Results & What They Mean

Speech Script:

"Now let's look at the results! This table shows all three brains and their scores. Lower RMSE is better, and you can see that the **Residual MLP won with a score of 3.0591** - that's the best performance."

"The 1D-CNN came in second with 3.0970, and the Plain MLP was third with 3.2452. What's interesting is that the Residual MLP achieved this with only 36,638 parameters - that's the brain size - while the CNN used over 200,000 parameters but still performed worse."

SLIDE 4

"If you look at this bar chart, you can see the comparison clearly. The Residual MLP is the shortest bar, which is best.

SLIDE 5

And these training curves show how the Residual MLP learned much better than the Plain MLP. The blue line drops faster and goes lower, which means it learned more effectively."

"What this tells us is that skip connections really help. The CNN was good too, but it used five times more brain power. And the Plain MLP without skip connections was the worst performer. In simple terms, an RMSE of 3.06 means the model is off by about 3 units on average - which is pretty good for a first attempt at this problem."

SLIDE 6: CNN Results (Extra Slide)

Speech Script:

"Here's a closer look at the CNN results. The training curves show the loss decreasing steadily over 50 epochs, which means the model learned well over time. It started with high error and gradually improved."

"The predicted versus actual plot shows our predictions are close to the real values, which is good. The blue lines are the actual values, and the red dashed lines are our predictions - you can see they track each other reasonably well."

"But some channels are harder to predict than others. This bar chart shows the error per channel, and you can see some have much higher error. This tells us where we need to improve. Overall, the CNN did a solid job with an RMSE of 3.0970 - a solid baseline for comparison."

SLIDE 7: How to Apply This Elsewhere

Speech Script:

"This project isn't just about weather! The same pattern works everywhere: take a few measurements and use AI to guess the full picture. This is a fundamental approach in many fields."

"For example, in **weather forecasting**, we take weather station data and predict the full weather map. In **medical imaging**, we take a few MRI slices and reconstruct a complete 3D brain scan. In **satellite imaging**, we take pixels from space and create full ground images. In **photo restoration**, we take a blurry photo and produce a clear image. And in **audio processing**, we take a short audio clip and reconstruct the full song."

"For this project, the next steps would be: testing on real atmospheric data instead of synthetic, adding attention mechanisms so the model can focus on important parts, trying a U-Net architecture which is powerful for reconstruction, and eventually deploying this as a website app that anyone can use."

"In one sentence: I trained AI brains to guess the full atmosphere from few measurements, and found that skip connections help the most!"

"Thank you for listening! Week 2 is complete!" 🎉

